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Inventor(s)	KIM; Seonil

SERVER SUPPORTING DISTRIBUTED DATABASE STRUCTURE AND OPERATION METHOD THEREOF

Abstract

A server is provided. The server includes a communication circuit, memory, including one or more storage media, storing instructions, and a processor communicatively coupled to the communication circuit and the memory, wherein the instructions, when executed by the processor, cause the server to collect a timestamp value indicating a point in time at which data writing is completed, determine a member to belong to a read pool by using the timestamp value in each storage, configure a read pool list so as to include the member determined through the determination operation and store the read pool list in the memory, receive a read request from a client apparatus, select a storage to provide data in response to the read request, receive data through the communication circuit from the storage selected from the read pool list, and transmit the received data to the client apparatus through the communication circuit.

Inventors: **KIM; Seonil (Suwon-si, KR)**

Applicant: **Samsung Electronics Co., Ltd.** (Suwon-si, KR)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation application, claiming priority under 35 U.S.C. § 365 (c), of an International application No. PCT/KR2023/010177, filed on Jul. 17, 2023, which is based on and claims the benefit of a Korean patent application number 10-2022-0185245, filed on Dec. 27, 2022, in the Korean Intellectual Property Office, and of a Korean patent application number 10-2023-0010215, filed on Jan. 26, 2023, in the Korean Intellectual Property Office, the disclosure of each of which is incorporated by reference herein in its entirety.

BACKGROUND
1. Field
[0002] The disclosure relates to a server supporting a distributed database structure, and an operation method thereof.
2. Description of Related Art
[0003] A distributed database structure may include a primary node and a replica node. The primary node may record data in a storage of the primary node in response to a write request from a client, read data from a storage in response to a read request from the client, and provide the same to the client. The replica node may read data from a storage (e.g., replica storage) of the replica node in response to a read request from a client, and provide the same to the client. The primary node may perform an operation of providing data stored in its own storage (original storage) to replica nodes. For example, a process of recording, in the replica storages, data updated in the original storage through the write request from the client may be performed via data communication between the primary node and the replica nodes.
[0004] The above information is presented as background information only to assist with an understanding of the disclosure. No determination has been made, and no assertion is made, as to whether any of the above might be applicable as prior art with regard to the disclosure.
SUMMARY
[0005] A distributed database structure may be classified into fully synchronous scheme and asynchronous schemes.
[0006] The fully synchronous scheme may provide strong or strict consistency to data. For example, a primary node and all replica nodes can maintain consistent data. A client may read the up to date data from the original storage of one of the replica storages.
[0007] The asynchronous scheme may provide eventual consistency. A speed at which the replica node responds to a read request from the client may be comparatively fast. However, according to a replica node which has responded, data provided to the client may not the up to date value.
[0008] According to various embodiments of the disclosure, in the asynchronous scheme, a server can relatively increase the reliability of data to be return to the client in response to a read request from the client. The technical problems of the disclosure are not limited to the above-described technical problems, and other unmentioned technical problems can be clearly understood from the description below by those skilled in the art to which the disclosure belongs.

[0009] Aspects of the disclosure are to address at least the above-mentioned problems and/or disadvantages and to provide at least the advantages described below. Accordingly, an aspect of the disclosure is to provide a server supporting a distributed database structure, and an operation method thereof.
[0010] Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.
[0011] In accordance with an aspect of the disclosure, a server is provided. The server includes a communication circuit, memory, including one or more storage media, storing instructions, and a processor communicatively coupled to the communication circuit and the memory, wherein the instructions, when executed, by the processor, cause the server to collect, from an original storage and replica storages through the communication circuit, a timestamp value indicating a time point at which data writing is completed, determine, from among the replica storages, a member to belong to a read pool, together with the original storage, by using the timestamp value in each of the storages, which has been collected through the collection, configure a read pool list to include the member determined through determining, and store the read pool list in the memory, receive a read request from a client device through the communication circuit, select, from the read pool list, a storage to provide data in response to the read request, receive data from the storage selected from the read pool list through the communication circuit, and transmit the received data to the client device through the communication circuit.
[0012] The instructions, when executed by the processor, in case that a first confidence time value remaining after subtracting a sum of a designated time unit value and a second timestamp value collected from a first replica storage from a sum of a first timestamp value collected from the original storage and an update period value indicating a period of the collection operation is equal to or smaller than a designated threshold time value, further cause the server to include the first replica storage in the read pool list.
[0013] The instructions, when executed by the processor, in case that a second confidence time value remaining after subtracting a sum of the unit time value and a third timestamp value collected from a second replica storage from a sum of the first timestamp value and the update period value exceeds the threshold time value, further cause the server to exclude the second replica storage from the read pool list.
[0014] The instructions, when executed by the processor to, in case that a difference between a first timestamp value collected from the original storage and a second timestamp value collected from a first replica storage is equal to or smaller than a designated reference value, further cause the server to include the first replica storage in the read pool list.
[0015] The instructions, when executed by the processor to, in case that a difference between the first timestamp value and a third timestamp value collected from a second replica storage exceeds the reference value, further cause the server to exclude the second replica storage from the read pool list.
[0016] The instructions, when executed by the processor to, in case that a write request is received in the communication circuit, further cause the server to control the communication circuit to transmit the write request to the original storage.
[0017] In accordance with another aspect of the disclosure, a server is provided. The server includes a communication circuit, memory, including one or more storage media and an original storage, storing instructions, and a processor communicatively coupled to the communication circuit and the memory, wherein the instructions, when executed by the processor, cause the server to receive a write request from a client device through the communication circuit, record data corresponding to the write request in the original storage, update a timestamp value used to configure a read pool, which can provide data in response to a read request from the client device, with a timestamp value indicating a time point at which the recording is completed, and transmit the updated timestamp value to replica storages through the communication circuit.

[0018] The instructions, when executed by the processor, further cause the server to record a timestamp value in a log file indicating use details of the original storage.
[0019] The instructions, when executed by the processor, in case that a read request is received in the communication circuit, further cause the server to control the communication circuit to transmit the read request to a server which manages the read pool.
[0020] In accordance with another aspect of the disclosure, a method of operating a server is provided. The method includes collecting, from an original storage and replica storages through a communication circuit of the server, a timestamp value indicating a time point at which data writing is completed, determining, from among the replica storages, a member to belong to a read pool, together with the original storage, by using the timestamp value in each of the storages, which has been collected through the collection, configuring a read pool list to include the member determined through the determination operation, and storing the read pool list in memory of the server, receiving a read request from a client device through the communication circuit, selecting, from the read pool list, a storage to provide data in response to the read request, receiving data from the storage selected from the read pool list through the communication circuit, and transmitting the received data to the client device through the communication circuit.
[0021] The configuring of the read pool list includes, in case that a first confidence time value remaining after subtracting a sum of a designated time unit value and a second timestamp value collected from a first replica storage from a sum of a first timestamp value collected from the original storage and an update period value indicating a period of the collection is equal to or smaller than a designated threshold time value, including the first replica storage in the read pool list.
[0022] The configuring of the read pool list includes, in case that a second confidence time value remaining after subtracting a sum of the unit time value and a third timestamp value collected from a second replica storage from a sum of the first timestamp value and the update period value exceeds the threshold time value, excluding the second replica storage from the read pool list.
[0023] The configuring of the read pool list includes, in case that a difference between a first timestamp value collected from the original storage and a second timestamp value collected from a first replica storage is equal to or smaller than a designated reference value, including the first replica storage in the read pool list.
[0024] The configuring of the read pool list includes, in case that a difference between the first timestamp value and a third timestamp value collected from a second replica storage exceeds the reference value, excluding the second replica storage from the read pool list.
[0025] In accordance with another aspect of the disclosure, a method of operating a server is provided. The method includes receiving a write request from a client device through a communication circuit of the server, recording data corresponding to the write request in an original storage of the server, updating a timestamp value used to configure a read pool, which can provide data in response to a read request from the client device, with a timestamp value indicating a time point at which the recording is completed, and transmitting the updated timestamp value to replica storages through the communication circuit.
[0026] A server according to various embodiments can provide a client device with a faster response and reliable data to a read request by distributing a load to the read request to replica storages. In addition, various effects directly or indirectly identified through the disclosure can be provided.
[0027] In accordance with another aspect of the disclosure, one or more non-transitory computer-readable storage media storing one or more computer programs including computer-executable instructions that, when executed by one or more processors of a server individually or collectively, cause the electronic device to perform operations are provided. The operations include collecting, from an original storage and replica storages through a communication circuit of the server, a timestamp value indicating a time point at which data writing is completed, determining, from

among the replica storages, a member to belong to a read pool, together with the original storage, by using the timestamp value in each of the storages, which has been collected through the collection, configuring a read pool list to include the member determined through the determination, and storing the read pool list in memory of the server, receiving a read request from a client device through the communication circuit, selecting, from the read pool list, a storage to provide data in response to the read request, receiving data from the storage selected from the read pool list through the communication circuit, and transmitting the received data to the client device through the communication circuit.
[0028] Other aspects, advantages, and salient features of the disclosure will become apparent to those skilled in the art from the following detailed description, which, taken in conjunction with the annexed drawings, discloses various embodiments of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0029] The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:
[0030] FIG. **1** is a block diagram illustrating an electronic device in a network environment according to an embodiment of the disclosure;
[0031] FIG. **2** illustrates a configuration of a distributed database system according to an embodiment of the disclosure;
[0032] FIG. **3** illustrates an operation performed by a data management server in response to a write request from a client device according to an embodiment of the disclosure;
[0033] FIG. **4** illustrates an operation performed by a proxy server in response to a read request from a client device according to an embodiment of the disclosure;
[0034] FIG. **5** is a flowchart illustrating operations of a second processor in a proxy server according to an embodiment of the disclosure; and
[0035] FIG. **6** is a flowchart illustrating operations of a first processor in a data management server according to an embodiment of the disclosure.
[0036] Throughout the drawings, it should be noted that like reference numbers are used to depict the same or similar elements, features, and structures.
DETAILED DESCRIPTION
[0037] The following description with reference to the accompanying drawings is provided to assist in a comprehensive understanding of various embodiments of the disclosure as defined by the claims and their equivalents. It includes various specific details to assist in that understanding but these are to be regarded as merely exemplary. Accordingly, those of ordinary skill in the art will recognize that various changes and modifications of the various embodiments described herein can be made without departing from the scope and spirit of the disclosure. In addition, descriptions of well-known functions and constructions may be omitted for clarity and conciseness.
[0038] The terms and words used in the following description and claims are not limited to the bibliographical meanings, but, are merely used by the inventor to enable a clear and consistent understanding of the disclosure. Accordingly, it should be apparent to those skilled in the art that the following description of various embodiments of the disclosure is provided for illustration purpose only and not for the purpose of limiting the disclosure as defined by the appended claims and their equivalents.
[0039] It is to be understood that the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a component surface” includes reference to one or more of such surfaces.

[0040] It should be appreciated that the blocks in each flowchart and combinations of the flowcharts may be performed by one or more computer programs which include computer-executable instructions. The entirety of the one or more computer programs may be stored in a single memory device or the one or more computer programs may be divided with different portions stored in different multiple memory devices.
[0041] Any of the functions or operations described herein can be processed by one processor or a combination of processors. The one processor or the combination of processors is circuitry performing processing and includes circuitry like an application processor (AP, e.g., a central processing unit (CPU)), a communication processor (CP, e.g., a modem), a graphical processing unit (GPU), a neural processing unit (NPU) (e.g., an artificial intelligence (AI) chip), a wireless-fidelity (Wi-Fi) chip, a Bluetooth™ chip, a global positioning system (GPS) chip, a near field communication (NFC) chip, connectivity chips, a sensor controller, a touch controller, a finger-print sensor controller, a display drive integrated circuit (IC), an audio CODEC chip, a universal serial bus (USB) controller, a camera controller, an image processing IC, a microprocessor unit (MPU), a system on chip (SoC), an IC, or the like.
[0042] FIG. **1** is a block diagram illustrating an electronic device in a network environment according to an embodiment of the disclosure.
[0043] Referring to FIG. **1**, an electronic device **101** in a network environment **100** may communicate with an external electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an external electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment of the disclosure, the electronic device **101** may communicate with the external electronic device **104** via the server **108**. According to an embodiment of the disclosure, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments of the disclosure, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments of the disclosure, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).
[0044] The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to one embodiment of the disclosure, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment of the disclosure, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.
[0045] The auxiliary processor **123** may control at least some of functions or states related to at

least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., a sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment of the disclosure, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment of the disclosure, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.
[0046] The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.
[0047] The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.
[0048] The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).
[0049] The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment of the disclosure, the receiver may be implemented as separate from, or as part of the speaker.
[0050] The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment of the disclosure, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.
[0051] The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment of the disclosure, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., the external electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.
[0052] The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment of the disclosure, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration

sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.
[0053] The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the external electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment of the disclosure, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.
[0054] A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the external electronic device **102**). According to an embodiment of the disclosure, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).
[0055] The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment of the disclosure, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.
[0056] The camera module **180** may capture a still image or moving images. According to an embodiment of the disclosure, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.
[0057] The power management module **188** may manage power supplied to the electronic device **101**. According to one embodiment of the disclosure, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).
[0058] The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment of the disclosure, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.
[0059] The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the external electronic device **102**, the external electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment of the disclosure, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN)). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.
[0060] The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR

access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the external electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment of the disclosure, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

[0061] The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment of the disclosure, the antenna module **197** may include an antenna including a radiating element including a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment of the disclosure, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment of the disclosure, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

[0062] According to various embodiments of the disclosure, the antenna module **197** may form a mmWave antenna module. According to an embodiment of the disclosure, the mmWave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

[0063] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

[0064] According to an embodiment of the disclosure, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the external electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment of the disclosure, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102** or **104**, or the server **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment of the disclosure, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment of the disclosure, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., a smart home, a smart city, a smart car, or healthcare) based on 5G communication technology or IoT-related technology.

[0065] FIG. **2** illustrates a configuration of a distributed database system **200** according to an embodiment of the disclosure. FIG. **3** illustrates an operation performed by a data management server in response to a write request from a client device according to an embodiment of the disclosure. FIG. **4** illustrates an operation performed by a proxy server in response to a read request from a client device according to an embodiment of the disclosure.

[0066] Referring to FIG. **2**, a system **200** may include a client device **210**, a data management server (e.g., primary node) **220**, a proxy server (e.g., proxy node) **230**, and replica storages (e.g., replica nodes) **240**.

[0067] A network **250** (e.g., the second network **199** of FIG. **1**) may include a network and/or a cellular network for remote communication, such as Internet or a computer network (e.g., LAN or WAN). The connection among the above-described devices **210**, **220**, **230**, and **240** in the system **200** may be established through the network **250**.

[0068] The client device **210** (e.g., the electronic device **101** of FIG. **1**) may perform data communication with the servers **220** and **230** through the network **250**. For example, referring to FIGS. **3** and **4**, the client device **210** may transmit a message (hereinafter, referred to as a write request) **310** requesting data writing or a message (hereinafter, referred to as a read request) **410** requesting data reading to the data management server **220** or the proxy server **230** through the network **250**.

[0069] According to an embodiment of the disclosure, the data management server **220** may include a timestamp management module **221**, a first communication circuit **227**, first memory **228**, and a first processor **229**. The first communication circuit **227**, the first memory **228**, and the first processor **229** may have substantially the same implementation as and perform the same function as the communication module **190**, the memory **130**, and the processor **120** of FIG. **1**, respectively.

[0070] When the read request is received in the first communication circuit **227**, the first processor **229** may control the first communication circuit **227** to transfer the read request to the proxy server **230**.

[0071] The first processor **229** may receive the write request from the client device **210** through the first communication circuit **227**. The first processor **229** may perform transaction in response to the write request from the client device **210**. For example, the first processor **229** may update data stored in an original storage **225** of the first memory **228** with new data corresponding to the write request. Referring to FIG. **3**, the first processor **229** may transmit a response message (e.g., an update complete message) **320** indicating whether the transaction has been successfully performed to the client device **210** through the first communication circuit **227**.

[0072] The first processor **229** may generate identification information (e.g., global transaction identification (GTID)) for identifying the performed transaction in response to the write request, and record the same in a log file (e.g., binary log) in which use details of the original storage **225** are recorded. The first processor **229** may identify data corresponding to the identification information in the original storage **225**, replicate the identified data, and transmit a replica to the replica storages **240**. Accordingly, the replica storages **240** may be synchronized with the original storage **225**.

[0073] According to an embodiment of the disclosure, the timestamp management module **221** may record a timestamp indicating a time point at which the transaction is completed (e.g., a time point at which updating is completed) in a log file of the original storage **225** together with the corresponding identification information. For example, the timestamp management module **221** may update the timestamp value recorded in the log file with a timestamp value indicating a time point at which the most recent transaction is completed. The timestamp management module **221** may transmit the replica to the replica storages **240** together with the updated timestamp value. Accordingly, a newly updated timestamp value may be recorded in a log file of each of the replica storages **240**. The timestamp management module **221** may include instructions which may be stored in the first memory **228** and executed by the first processor **229**.

[0074] Referring to FIG. **3**, the timestamps recorded in the original storage **225**, a first replica storage **301**, and a third replica storage **303** may be all "20" and identical to one another. When the timestamps are identical, it may mean that the first replica storage **301** and the third replica storage **303** are synchronized with the original storage **225**, and accordingly, the replica stored in the first replica storage **301** and the replica stored in the third replica storage **303** are reliable. Upon the delay in the transmission to the second replica storage **302**, the timestamp recorded in the second replica storage **302** may have a value of "18", which is smaller than the value of the timestamp recorded in the original storage **225**. When the second replica storage **302** has the timestamp having the value smaller than the timestamp of the original storage **225**, it may mean that there is high possibility that the second replica storage **302** has not been synchronized with the original storage **225**.

[0075] According to an embodiment of the disclosure, the proxy server **230** may include a read pool management module **231**, a second communication circuit **237**, second memory **238**, and a second processor **239**. The second communication circuit **237**, the second memory **238**, and the second processor **239** may have substantially the same implementation as and perform the same function as the communication module **190**, the memory **130**, and the processor **120** of FIG. **1**, respectively.

[0076] According to an embodiment of the disclosure, the read pool management module **231** may identify the original storage **225** and the replica storages **240** at each determine period Δt so as to collect timestamp values of the respective storages. The read pool management module **231** may use the collected timestamp values to select, from among the replica storages **240**, a member to belong to a read pool (in other words, a reliable storage group) which can provide data to the client device **210**. The original storage **225** may provide a source of the replica, and thus the read pool management module **231** may determine the original storage **225** as a member of the read pool which can provide data to the client device **210**.

[0077] According to an embodiment of the disclosure, the read pool management module **231** may determine, based on a difference between the timestamp value collected from the original storage **225** and the timestamp value collected from the replica storage, whether the replica storage can be a member of the read pool. For example, when the difference has a value equal to or smaller than a designated reference value, the read pool management module **231** may determine the replica storage as a member of the read pool. When the difference exceeds the reference value, the read pool management module **231** may exclude the replica storage from the read pool.

[0078] According to an embodiment of the disclosure, the read pool management module **231** may use the timestamp value collected from the original storage **225** and the timestamp value collected from the replica storage, so as to calculate a confidence time value, and may determine, based on the calculated confidence time value, whether the replica storage can be a member of the read pool.

For the calculation of the confidence time value, for example, the following Equation 1 can be used. When the confidence time value calculated using Equation 1 is equal to or smaller than a designated threshold time value, the read pool management module **231** may determine the replica storage as a member of the read pool. When the calculated confidence time value exceeds the threshold time value, the read pool management module **231** may exclude the replica storage from the read pool.

(Timestamp value of original storage 225+update period value (Δt))−(unit time value+timestamp value of replica storage) Equation 1

[0079] Equation 1 above is merely provided as an example to assist understanding, and the disclosure is not limited thereto, and alternation, application, or extension can be made in various schemes.

[0080] An example of a method for updating a read pool list **235** by using Equation 1 is described with reference to Table 1, Table 2, and Table 3. For example, the updating process is described by configuring the threshold time value as 5 seconds, configuring the update period value Δt as 3 seconds, and configuring the unit time value as 1 second.

TABLE-US-00001 TABLE 1 Primary Replica 1 Replica 2 Replica 3 Replica 4 Time timestamp timestamp timestamp timestamp timestamp Read pool 0 0 0 0 0 0 Primary

[0081] The initial timestamp values of all storages (Primary, Replica 1, Replica 2, Replica 3, and Replica 4) may be all "0" as shown in Table 1, and the read pool may include an original (primary) storage only.

TABLE-US-00002 TABLE 2 Primary Replica 1 Replica 2 Replica 3 Replica 4 Time timestamp timestamp timestamp timestamp timestamp Read pool 0 0 0 0 0 0 Primary 0 + Δt 3 3 1 1 0 Primary Replica 1 Replica 2 Replica 3 Replica 4 0 + Δt*2 6 5 3 1 0 Primary Replica 1 Replica 2

[0082] Table 2 above indicates the updating has been performed two times. When the timestamp value is first updated (e.g., when time is 0+Δt), transmission delay may occur in Replica 2, 3, and 4 among the replica storages. During primary updating, Replica 1, 2, 3, and 4 may all satisfy a condition to belong to the read pool as shown in the calculation result below.

[00001]Replica1: (3 + 3) - (1 + 3) = 2 <= 5Replica2: (3 + 3) - (1 + 1) = 4 <= 5
Replica3: (3 + 3) - (1 + 1) = 4 <= 5Replica4: (3 + 3) - (1 + 0) = 5 <= 5

[0083] During secondary updating, Replica 3 and 4 may fail to satisfy the condition and may be excluded from the read pool.

[00002]Replica1: (6 + 3) - (1 + 5) = 3 <= 5Replica2: (6 + 3) - (1 + 3) = 5 <= 5
Replica3: (6 + 3) - (1 + 1) = 7 > 5Replica4: (6 + 3) - (1 + 0) = 8 > 5

[0084] During the secondary updating, there is a 5-second difference between the timestamp of Replica 2 and the timestamp of the primary storage, this difference is identical to a threshold time value, and thus is can be considered that Replica 3 belongs to the read pool. However, based on a 3-second update period, it can be more appropriate to exclude Replica 3 from the read pool. For example, when the timestamp of Replica 3 is not updated at the next period, a difference between the timestamp value of Replica 3 and the timestamp value of the original (primary) timestamp may exceed a threshold time value before the next period arrives. The following Table 3 can be referred to.

TABLE-US-00003 TABLE 3 Primary Replica 1 Replica 2 Replica 3 Replica 4 Time timestamp timestamp timestamp timestamp timestamp Read pool 0 0 0 0 0 0 Primary 0 + Δt 3 3 1 1 0 Primary Replica 1 Replica 2 Replica 3 0 + Δ t*2 6 5 3 1 0 Primary 0 + Δ t*2 + 1 7 5 5 1 0 Replica 1

[0085] According to an embodiment of the disclosure, the timestamp management module **221** may generate, regardless of whether a transaction occurs, a dummy request at each unit time (1 second) so as to update a timestamp value of the original (primary) storage. For example, referring to Table 2, when 1 second (unit time) passes after the secondary updating, the timestamp value of the original (primary) storage is updated from 6 seconds to 7 seconds, and the timestamp value of

Replica 3 may not be updated. Accordingly, a difference between the timestamp value of Replica 3 and the timestamp value of the original (primary) timestamp becomes 6 seconds. This difference exceeds the threshold time value, which corresponds to a case where Replica 3 needs to be excluded form the read pool. Accordingly, it can be appropriate to include the update period value Δt to the equation used to update the read pool.
[0086] The read pool management module **231** may store the read pool list **235** in the second memory **238** and update the read pool list **235** at each designated period Δt.
[0087] The read pool management module **231** may include instructions which may be stored in the second memory **238** and executed by the second processor **239**.
[0088] When a write request is received in the second communication circuit **237**, the second processor **239** may control the second communication circuit **237** to transmit the write request to the data management server **220**.
[0089] The second processor **239** may receive a read request from the client device **210** through the second communication circuit **237**. The second processor **239** may identify the read pool list **235** in response to the read request, and may select, from the read pool list **235**, a storage to provide data to the client device **210**.
[0090] Referring to FIG. **4**, the second processor **239** may transmit, from the read pool **430** to the selected storage through the second communication circuit **237**, a message **420** requesting to provide data. The second processor **239** may receive data from the selected storage and transmit the received data to the client device **210** as a response **440** to the read request **410**.
[0091] The terms “first” and “second,” as used above and as will be used hereinafter, are merely used for distinguishing between names, and no special meaning, such as importance or order is assigned thereto.
[0092] FIG. **5** is a flowchart illustrating operations of a second processor in a proxy server according to an embodiment of the disclosure.
[0093] Referring to FIG. **5**, in an embodiment below, respective operations may be subsequently performed, but are not necessarily performed subsequently. For example, the sequence of the respective operations may be changed, and two or more operations may be performed in parallel.
[0094] According to an embodiment of the disclosure, operations **510** to **570** may be understood to be performed by a processor (e.g., the second processor **239** of FIG. **2**) of a proxy server (e.g., the proxy server **230** of FIG. **2**).
[0095] In operation **510**, the second processor **239** may collect a timestamp value at each designated period Δt from an original storage **225** and replica storages **240** through a second communication circuit **237**.
[0096] In operation **520**, the second processor **239** may determine a member to belong to a read pool together with the original storage **225** among the replica storages **240** by using the collected timestamp value of each storage. In operation **530**, the second processor **239** may configure a read pool list **235** to include the determined member and store the same in second memory **238**.
[0097] According to an embodiment of operation **530**, when a first confidence time value remaining after subtracting a sum of a unit time value (e.g., 1 second) and a second timestamp value collected from a first replica storage from a sum of a first timestamp value collected from the original storage **225** and an update period value (e.g., 3 seconds) is equal to smaller than a threshold time value (e.g., 5 seconds), the second processor **239** may include the first replica storage in the read pool list **235**. When a second confidence time value remaining after subtracting a sum of the unit time value and a third timestamp value collected from a second replica storage from a sum of the first timestamp value and the update period value exceeds a threshold time value, the second processor **239** may exclude the second replica storage from the read pool list **235**.
[0098] According to an embodiment of operation **530**, when a difference between the first timestamp value and the second timestamp value has a value equal to or smaller than a designated reference value, the second processor **239** may include the first replica storage in the read pool list

235. When a difference between the first timestamp value and a third timestamp value has a value exceeding a reference value, the second processor **239** may exclude the second replica storage from the read pool list **235**.
[0099] In operation **540**, the second processor **239** may receive a read request from a client device **210** through the second communication circuit **237**. In operation **550**, the second processor **239** may select a storage to provide data from the read pool list **235** in response to the read request. The selection scheme varies, and for example, the storage can be randomly selected from the read pool.
[0100] In operation **560**, the second processor **239** may receive data from the selected storage through the second communication circuit **237**.
[0101] In operation **570**, the second processor **239** may transmit data received from the selected storage to the client device **210** through the second communication circuit **237**.
[0102] FIG. **6** is a flowchart illustrating operations of a first processor in a data management server according to an embodiment of the disclosure.
[0103] Referring to FIG. **6**, in an embodiment below, respective operations may be subsequently performed, but are not necessarily performed subsequently. For example, the sequence of the respective operations may be changed, and two or more operations may be performed in parallel.
[0104] According to an embodiment of the disclosure, operations **610** to **640** may be understood to be performed by a processor (e.g., the first processor **229** of FIG. **2**) of a data management server (e.g., the data management server **220** of FIG. **2**).
[0105] In operation **610**, the first processor **229** may receive a write request from a client device **210** through a first communication circuit **227**.
[0106] In operation **620**, the first processor **229** may record data corresponding to the write request in an original storage **225** (e.g., update existing data with new data corresponding to the write request).
[0107] In operation **630**, the first processor **229** may update a timestamp value stored in the original storage **225** with a timestamp value indicating a time point at which the recording is completed.
[0108] In operation **640**, the first processor **229** may transmit the updated timestamp value to the replica storages **240** through the first communication circuit **227**.
[0109] According to an embodiment of the disclosure, a server (e.g., the proxy server **230**) may include a communication circuit, a processor connected to the communication circuit, and memory connected to the processor. The memory may store instructions which, when executed, cause the processor to perform a collection operation of collecting, from an original storage and replica storages through the communication circuit, a timestamp value indicating a time point at which data writing is completed. The instructions may cause the processor to perform a determination operation of determining, from among the replica storages, a member to belong to a read pool together with the original storage by using the timestamp value in each storage, which has been collected through the collection operation. The instructions may cause the processor to configure a read pool list so that the member determined through the determination operation is included and store the read pool list in the memory. The instructions may cause the processor to receive a read request from a client device through the communication circuit. The instructions may cause the processor to select, from the read pool list, a storage to provide data in response to the read request. The instructions may cause the processor to receive data from the storage selected from the read pool list through the communication circuit. The instructions may cause the processor to transmit the received data to the client device through the communication circuit.
[0110] The instructions may cause the processor to, in case that a first confidence time value remaining after subtracting a sum of a designated time unit value and a second timestamp value collected from a first replica storage from a sum of a first timestamp value collected from the original storage and an update period value indicating a period of the collection operation is equal to or smaller than a designated threshold time value, cause the first replica storage to be included in the read pool list.

[0111] The instructions may cause the processor to, in case that a second confidence time value remaining after subtracting a sum of the unit time value and a third timestamp value collected from a second replica storage from a sum of the first timestamp value and the update period value exceeds the threshold time value, exclude the second replica storage from the read pool list.
[0112] The instructions may cause the processor to, in case that a difference between a first timestamp value collected from the original storage and a second timestamp value collected from a first replica storage has a value equal to or smaller than a reference value, cause the first replica storage to be included in the read pool list.
[0113] The instructions may cause the processor to, in case that a difference between the first timestamp value and a third timestamp value collected from a second replica storage exceeds the reference value, exclude the second replica storage from the read pool list.
[0114] The instructions may cause the processor to, in case that a write request is received in the communication circuit, control the communication circuit to transmit the write request to the original storage.
[0115] According to an embodiment of the disclosure, a server (e.g., the data management server **220**) may include a communication circuit, a processor connected to the communication circuit, and memory connected to the processor and including an original storage. The memory may store instructions which, when executed, cause the processor to receive a write request from a client device through the communication circuit. The memory may store instructions which, when executed, cause the processor to record data corresponding to the write request in the original storage. The instructions may cause the processor to update a timestamp value used to configure a read pool which can provide data in response to a read request from the client device with a timestamp value indicating a time point at which the recording is completed. The instructions may cause the processor to transmit the updated timestamp value to replica storages through the communication circuit.
[0116] The instructions may cause the processor to record a timestamp value in a log file indicating use details of the original storage.
[0117] The instructions may cause the processor to, in case that a read request is received in the communication circuit, control the communication circuit to transmit the read request to a server which manages the read pool.
[0118] According to an embodiment of the disclosure, a method of operating a server (e.g., the proxy server **230**) is provided. The method may include collecting, from an original storage and replica storages through a communication circuit of the server, a timestamp value indicating a time point at which data writing is completed (e.g., operation **510**). The method may include determining, from among the replica storages, a member to belong to a read pool together with the original storage by using the timestamp value in each storage, which has been collected through the collection (e.g., operation **520**). The method may include configuring a read pool list so that the member determined through the determination operation is included and storing the read pool list in memory of the server (e.g., operation **530**). The method may include receiving a read request from a client device through the communication circuit (e.g., operation **540**). The method may include selecting, from the read pool list, a storage to provide data in response to the read request (e.g., operation **550**). The method may include receiving data from the storage selected from the read pool list through the communication circuit (e.g., operation **560**). The method may include transmitting the received data to the client device through the communication circuit (e.g., operation **570**).
[0119] The configuring of the read pool list may include, in case that a first confidence time value remaining after subtracting a sum of a designated time unit value and a second timestamp value collected from a first replica storage from a sum of a first timestamp value collected from the original storage and an update period value indicating a period of the collection is equal to or smaller than a designated threshold time value, causing the first replica storage to be included in

the read pool list.
[0120] The configuring of the read pool list may include, in case that a second confidence time value remaining after subtracting a sum of the unit time value and a third timestamp value collected from a second replica storage from a sum of the first timestamp value and the update period value exceeds the threshold time value, excluding the second replica storage from the read pool list.
[0121] The configuring of the read pool list may include, in case that a difference between a first timestamp value collected from the original storage and a second timestamp value collected from a first replica storage has a value equal to or smaller than a reference value, causing the first replica storage to be included in the read pool list.
[0122] The configuring of the read pool list may include, in case that a difference between the first timestamp value and a third timestamp value collected from a second replica storage exceeds the reference value, excluding the second replica storage from the read pool list.
[0123] According to an embodiment of the disclosure, a method of operating a server (e.g., the data management server **220**) is provided. The method may include receiving a write request from a client device through a communication circuit of the server (e.g., operation **610**). The method may include recording data corresponding to the write request in an original storage of the server (e.g., operation **620**). The method may include updating a timestamp value used to configure a read pool which can provide data in response to a read request from the client device with a timestamp value indicating a time point at which the recording is completed (e.g., operation **630**). The method may include transmitting the updated timestamp value to replica storages through the communication circuit (e.g., operation **640**).
[0124] The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.
[0125] It should be appreciated that various embodiments of the disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.
[0126] As used in connection with various embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment of the disclosure, the module may be implemented in a form of an application-specific integrated circuit (ASIC).
[0127] Various embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., internal memory **136** or external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may

invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a complier or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

[0128] According to an embodiment of the disclosure, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

[0129] According to various embodiments of the disclosure, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments of the disclosure, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments of the disclosure, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments of the disclosure, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

[0130] It will be appreciated that various embodiments of the disclosure according to the claims and description in the specification can be realized in the form of hardware, software or a combination of hardware and software.

[0131] Any such software may be stored in non-transitory computer readable storage media. The non-transitory computer readable storage media store one or more computer programs (software modules), the one or more computer programs include computer-executable instructions that, when executed by one or more processors of an electronic device, cause the electronic device to perform a method of the disclosure.

[0132] Any such software may be stored in the form of volatile or non-volatile storage, such as, for example, a storage device like read only memory (ROM), whether erasable or rewritable or not, or in the form of memory, such as, for example, random access memory (RAM), memory chips, device or integrated circuits or on an optically or magnetically readable medium, such as, for example, a compact disk (CD), digital versatile disc (DVD), magnetic disk or magnetic tape or the like. It will be appreciated that the storage devices and storage media are various embodiments of non-transitory machine-readable storage that are suitable for storing a computer program or computer programs comprising instructions that, when executed, implement various embodiments of the disclosure. Accordingly, various embodiments provide a program comprising code for implementing apparatus or a method as claimed in any one of the claims of this specification and a non-transitory machine-readable storage storing such a program.

[0133] While the disclosure has been shown and described with reference to various embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the spirit and scope of the disclosure as defined by the appended claims and their equivalents.

Claims

1. A server comprising: a communication circuit; memory, comprising one or more storage media, storing instructions; and a processor communicatively coupled to the communication circuit and the memory, wherein the instructions, when executed by the processor, cause the server to: collect, from an original storage and replica storages through the communication circuit, a timestamp value indicating a time point at which data writing is completed, determine, from among the replica storages, a member to belong to a read pool, together with the original storage, by using the timestamp value in each of the storages, which has been collected through the collection, configure a read pool list to include the member determined through determining, and store the read pool list in the memory, receive a read request from a client device through the communication circuit, select, from the read pool list, a storage to provide data in response to the read request, receive data from the storage selected from the read pool list through the communication circuit, and transmit the received data to the client device through the communication circuit.
2. The server of claim 1, wherein the instructions, when executed by the processor, in case that a first confidence time value remaining after subtracting a sum of a designated time unit value and a second timestamp value collected from a first replica storage from a sum of a first timestamp value collected from the original storage and an update period value indicating a period of the collection is equal to or smaller than a designated threshold time value, further cause the server to include the first replica storage in the read pool list.
3. The server of claim 2, wherein the instructions, when executed by the processor, in case that a second confidence time value remaining after subtracting a sum of the unit time value and a third timestamp value collected from a second replica storage from the sum of the first timestamp value and the update period value exceeds the threshold time value, further cause the server to exclude the second replica storage from the read pool list.
4. The server of claim 1, wherein the instructions, when executed by the processor, in case that a difference between a first timestamp value collected from the original storage and a second timestamp value collected from a first replica storage is equal to or smaller than a designated reference value, further cause the server to include the first replica storage in the read pool list.
5. The server of claim 4, wherein the instructions, when executed by the processor, in case that a difference between the first timestamp value and a third timestamp value collected from a second replica storage exceeds the reference value, further cause the server to exclude the second replica storage from the read pool list.
6. The server of claim 1, the instructions, when executed by the processor, in case that a write request is received via the communication circuit, further cause the server to control the communication circuit to transmit the write request to the original storage.
7. A server comprising: a communication circuit; memory, comprising one or more storage media and an original storage, storing instructions; and a processor communicatively coupled to the communication circuit and the memory, wherein the instructions, when executed by the processor, cause the server to: receive a write request from a client device through the communication circuit, record data corresponding to the write request in the original storage, update a timestamp value used to configure a read pool, which can provide data in response to a read request from the client device, with a timestamp value indicating a time point at which the recording is completed, and transmit the updated timestamp value to replica storages through the communication circuit.
8. The server of claim 7, wherein the instructions, when executed by the processor, further cause

the server to record a timestamp value in a log file indicating use details of the original storage.
9. The server of claim 7, wherein the instructions, when executed by the processor, in case that a read request is received via the communication circuit, further cause the server to control the communication circuit to transmit the read request to a server which manages the read pool.
10. A method of operating a server, the method comprising: collecting, from an original storage and replica storages through a communication circuit of the server, a timestamp value indicating a time point at which data writing is completed; determining, from among the replica storages, a member to belong to a read pool, together with the original storage, by using the timestamp value in each of the storages, which has been collected through the collection; configuring a read pool list to include the member determined through the determination, and storing the read pool list in memory of the server; receiving a read request from a client device through the communication circuit; selecting, from the read pool list, a storage to provide data in response to the read request; receiving data from the storage selected from the read pool list through the communication circuit; and transmitting the received data to the client device through the communication circuit.
11. The method of claim 10, wherein the configuring of the read pool list comprises, in case that a first confidence time value remaining after subtracting a sum of a designated time unit value and a second timestamp value collected from a first replica storage from a sum of a first timestamp value collected from the original storage and an update period value indicating a period of the collection is equal to or smaller than a designated threshold time value, including the first replica storage in the read pool list.
12. The method of claim 11, wherein the configuring of the read pool list comprises, in case that a second confidence time value remaining after subtracting a sum of the unit time value and a third timestamp value collected from a second replica storage from the sum of the first timestamp value and the update period value exceeds the threshold time value, excluding the second replica storage from the read pool list.
13. The method of claim 12, wherein the configuring of the read pool list comprises, in case that a difference between the first timestamp value collected from the original storage and the second timestamp value collected from the first replica storage is equal to or smaller than a designated reference value, including the first replica storage in the read pool list.
14. The method of claim 13, wherein the configuring of the read pool list comprises, in case that a difference between the first timestamp value and the third timestamp value collected from the second replica storage exceeds the reference value, excluding the second replica storage from the read pool list.
15. The method of claim 13, further comprising: in case that a write request is received via the communication circuit, controlling the communication circuit to transmit the write request to the original storage.
16. A method of operating a server, the method comprising: receiving a write request from a client device through a communication circuit of the server; recording data corresponding to the write request in an original storage of the server; updating a timestamp value used to configure a read pool, which can provide data in response to a read request from the client device, with a timestamp value indicating a time point at which the recording is completed; and transmitting the updated timestamp value to replica storages through the communication circuit.
17. The method of claim 16, further comprising: recording a timestamp value in a log file indicating use details of the original storage.
18. The method of claim 16, further comprising: in case that a read request is received via the communication circuit, controlling the communication circuit to transmit the read request to a server which manages the read pool.
19. One or more non-transitory computer-readable storage media storing one or more computer programs including computer-executable instructions that, when executed by one or more processors of a server individually or collectively, cause the server to perform operations, the

operations comprising: collecting, from an original storage and replica storages through a communication circuit of the server, a timestamp value indicating a time point at which data writing is completed; determining, from among the replica storages, a member to belong to a read pool, together with the original storage, by using the timestamp value in each of the storages, which has been collected through the collection; configuring a read pool list to include the member determined through the determination, and storing the read pool list in memory of the server; receiving a read request from a client device through the communication circuit; selecting, from the read pool list, a storage to provide data in response to the read request; receiving data from the storage selected from the read pool list through the communication circuit; and transmitting the received data to the client device through the communication circuit.

20. The one or more non-transitory computer-readable storage media of claim 19, wherein the configuring of the read pool list comprises, in case that a first confidence time value remaining after subtracting a sum of a designated time unit value and a second timestamp value collected from a first replica storage from a sum of a first timestamp value collected from the original storage and an update period value indicating a period of the collection is equal to or smaller than a designated threshold time value, including the first replica storage in the read pool list.